

Accounting, Analysis and Governance in Complex Deals

Arndt Weinrich

weinrich@rsm.nl

Rotterdam School of Management

Valuation and Pricing

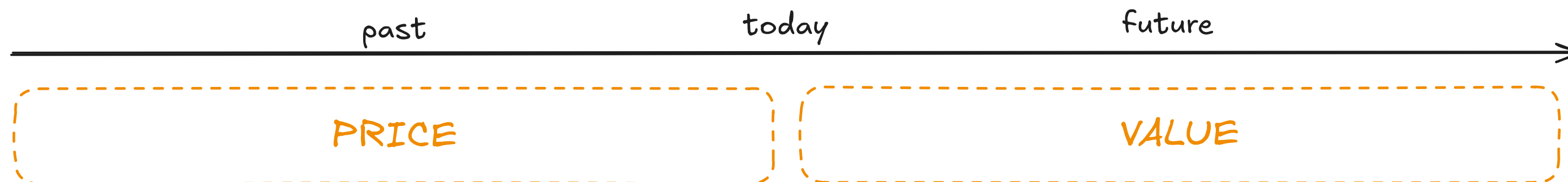
BMME303

Accounting, Analysis and Governance in Complex Deals

Value and Price

Setting the Stage

Value and Price Refer to Different Things



Price	Value
Observable/Unique	Can be Ambiguous
No Model	Model/Theory Needed

We will ignore this distinction and use the terms interchangeably

Measuring Value

Measuring Value

Valuation and Pricing

Under (restrictive) assumptions, multiple valuation methods collapse to the same result.

Let's explore when discount valuation approaches and ratio/multiple approaches can be seen as two sides of the same coin.

From Fundamental to Market-Based Valuation

Valuation and Pricing

Dividend Discount Model (DDM) w/o Growth

Assume that all earnings are paid out as dividends:

$$\boxed{P_0 = \frac{D_1}{r - g}} = \frac{E_1}{r} \Rightarrow P_0 = \frac{E_1}{r}$$

with $D = E$ and $g = 0$, where

- D_1 : Dividend next year, which equals earnings per share E_1
- r : Cost of Equity (required return)
- P_0 : Current stock price, which can be expressed as $\frac{E_1}{r}$ under DDM w/o growth

DDM is one of the most fundamental approaches to valuation. It states that the value of a stock today equals the present value of all future dividends. (Session 1 BM04AFM!)

From Fundamental to Market-Based Valuation

Valuation and Pricing

P/E Ratio

Rearrange $P_0 = \frac{E_1}{r}$ from DDM w/o growth to derive at a forward looking P/E ratio:

$$\frac{P_0}{E_1} = \frac{E_1/r}{E_1} \Rightarrow \boxed{P/E = \frac{1}{r}}$$

$\Rightarrow$ The forward-looking P/E ratio is the ratio of the current share price to next year's earnings

From Fundamental to Market-Based Valuation

Valuation and Pricing

P/E Ratio

P/E ratio determines a common **Multiples Approach** to valuation:

$$P_0 = E_1 \times (\text{Average of comparable firms' PE ratios})$$

⇒ **Multiples are fundamentally connected to discount models!**

- Incorporate the same fundamental valuation principles as more complex discount valuation approaches, but with greater simplicity.
- Selecting truly comparable firms is critical for the validity of the multiples approach!

From Fundamental to Market-Based Valuation

Valuation and Pricing

Equivalence of P/E and DDM

Multiples Approach

$$P_0 = E_1 \times (\text{Average of comparable firms' PE ratios})$$

DDM w/o Growth

$$P_0 = E_1 \times \frac{1}{r}$$

⇒ Both approaches yield the same valuation when the average P/E of comparable firms appropriately reflects required return (and growth) of the focal firm.

From Fundamental to Market-Based Valuation

Valuation and Pricing

Market-to-Book Ratio M/B

Derive at another direct approach to equity valuation:

- Define B as the Book value of Equity, then $ROE_1 = \frac{E_1}{B_0} \Rightarrow E_1 = ROE_1 \times B_0$
- $P_0 = \frac{E_1}{r} = \frac{ROE_1 \times B_0}{r}$

$$M/B = \frac{P_0}{B_0} = \frac{ROE_1 \times B_0 / r}{B_0} \Rightarrow \boxed{M/B = \frac{ROE}{r}}$$

$\Rightarrow$ **M/B incorporate profitability beyond just the required return (P/E)**

From Fundamental to Market-Based Valuation

Valuation and Pricing

Reconciliation between P/E and M/B

- $P/E = \frac{1}{r}$
- $M/B = \frac{ROE}{r} = ROE \times P/E$

$$M/B = ROE \times P/E$$

From Fundamental to Market-Based Valuation

Valuation and Pricing

Early Conclusions (No Growth Case)

- P/E ratio is entirely driven by the required return r , nothing else.
- M/B ratio is driven by both profitability (ROE) and the required return r .
- ROE vs. r is the key comparison for value creation:
 - If $\text{ROE} > r \Rightarrow \text{M/B} > 1$ (firm creates value)
 - If $\text{ROE} = r \Rightarrow \text{M/B} = 1$
 - If $\text{ROE} < r \Rightarrow \text{M/B} < 1$ (firm destroys value)

From Fundamental to Market-Based Valuation

Valuation and Pricing

What happens if we introduce growth?

	w/o Growth	w Growth	Note
P/E	$\frac{1}{r}$	$\frac{1-b}{r-ROE \times b}$	P/E increases only if growth is value-creating (ROE > r)
M/B	$\frac{ROE}{r}$	$\frac{ROE \times (1-b)}{r-ROE \times b}$	M/B is sensitive to both payout and profitability

⇒ **Growth introduces important tradeoffs between payout, reinvestment rate, and return on equity**

The growth rate g is defined as the product of the retention ratio $b = \frac{E-D}{E}$ and the return on equity (ROE). This means that the growth rate is directly influenced by how much of the earnings are retained for reinvestment in the firm and how effectively those retained earnings generate returns.

Early Takeaways

From Fundamental to Market-Based Valuation

Discount Rate r

- A **higher** r decreases both **P/E** and **M/B**
- Opportunity cost of capital for investors

Profitability: ROE

- If $ROE > r$: the firm creates value $\Rightarrow M/B > 1$.
- If $ROE < r$: the firm destroys value $\Rightarrow M/B < 1$.
- Also indirectly boosts P/E when retained earnings fuel value-creating growth.

Growth Rate g

- **P/E** and **M/B** both **increase** if retained earnings (via $b > 0$) fund profitable growth:
 - Only if **ROE** $> r$
 - Otherwise, growth **reduces value** (you reinvest at a rate lower than investor return expectations)

Payout Policy ($Payout\ Rate = 1 - b$)

- **Lower payouts** $\Rightarrow$ more retained earnings $\Rightarrow$ higher growth (if $ROE > r$)
- **Higher payouts** $\Rightarrow$ less growth but more cash returned to shareholders

From Fundamental to Market-Based Valuation

Valuation and Pricing

What happens if we want to value equity and debt?

Equity-side	Enterprise-side
Cost of Equity	WACC (Weighted Average Cost of Capital)
ROE	ROIC (Return on Invested Capital)
P/E	EV/EBIT or EV/EBIT(D)A
M/B	EV / IC (Invested Capital)

- Equity value = Value of shareholders' stake in the business
- Enterprise value = Value of the whole business, sum of the market values of the various claims on the business

Multiples in Action

Segway through the Valuation Jungle

DePamphilis (2025) *Relative, Asset-Oriented, and Real Option Valuation Basics*

Selected Valuation Approaches

- Comparable firms, transactions [...]
- Enterprise value (EV) to EBITDA
- Price / Earnings to Growth ratio (PEG)
- Value-Driver-Based Valuation
- [...]

Multiples Scheme in Shaffer (2024)

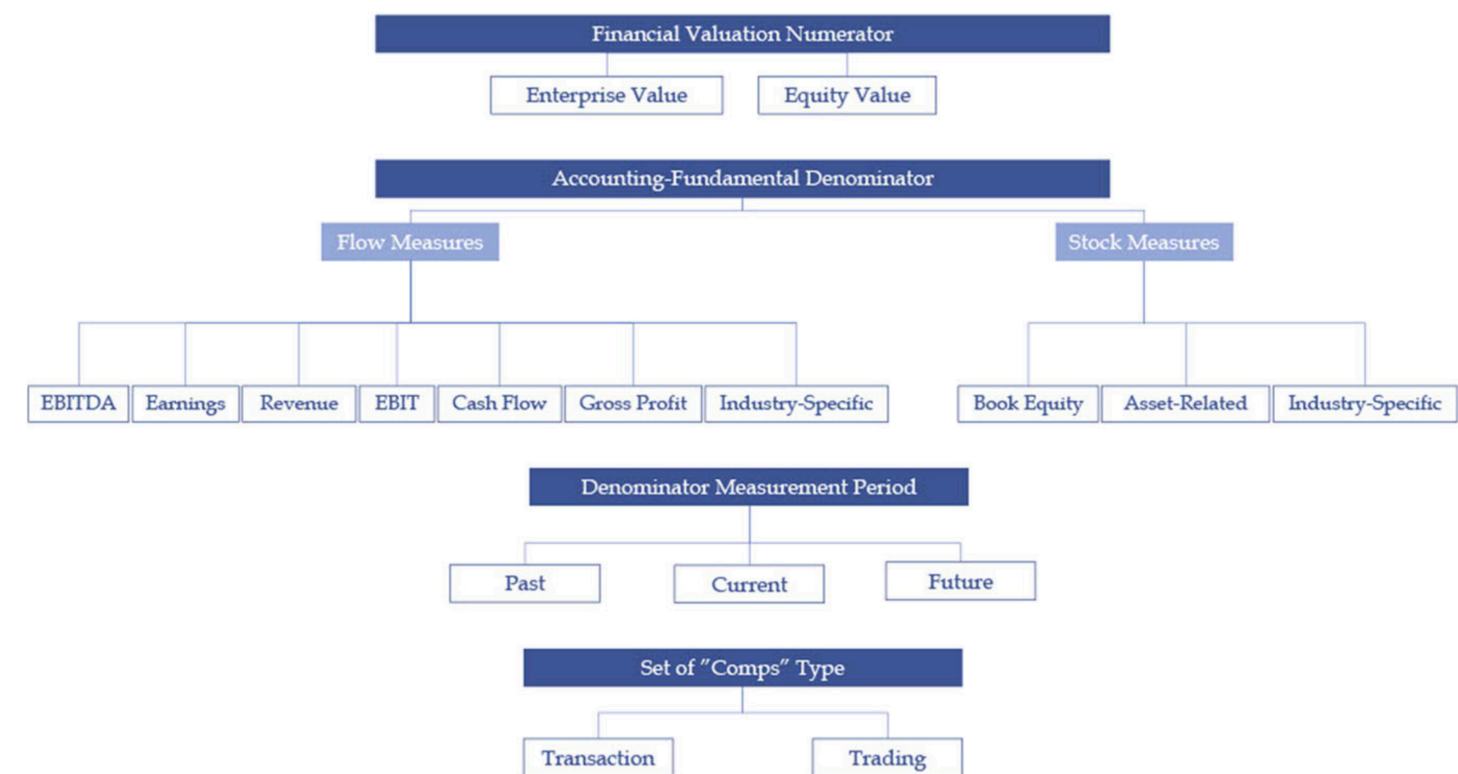


Fig. 1 Standardization scheme. Figure 1 illustrates the dimensions used to categorize the valuation ratios

Segway through the Valuation Jungle

Valuation and Pricing

Valuation Numerator

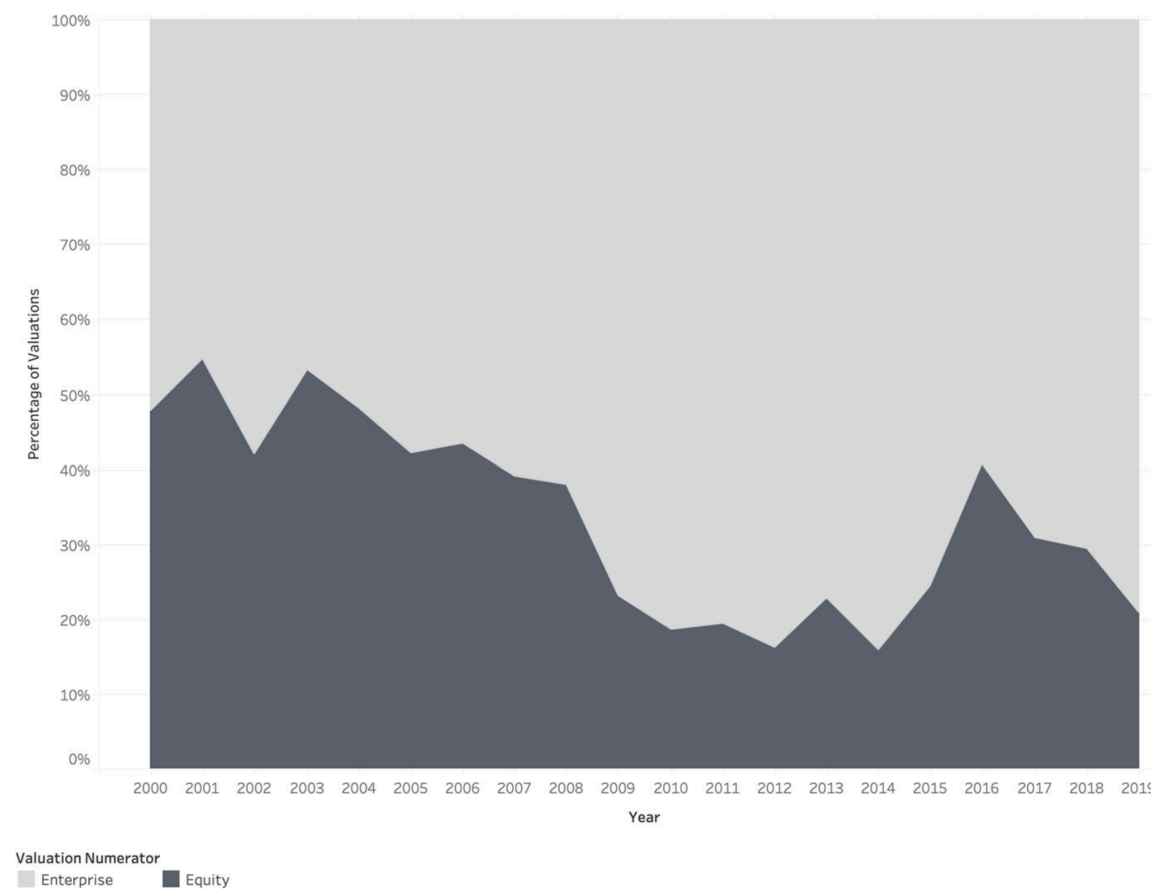


Fig. 2 Valuation numerator over time. Figure 2 plots the proportion of enterprise value valuation multiples versus direct equity value valuation multiples, from 2000 through 2019

Valuation **numerator** over time (Shaffer 2024)

Value-Driver Denominators

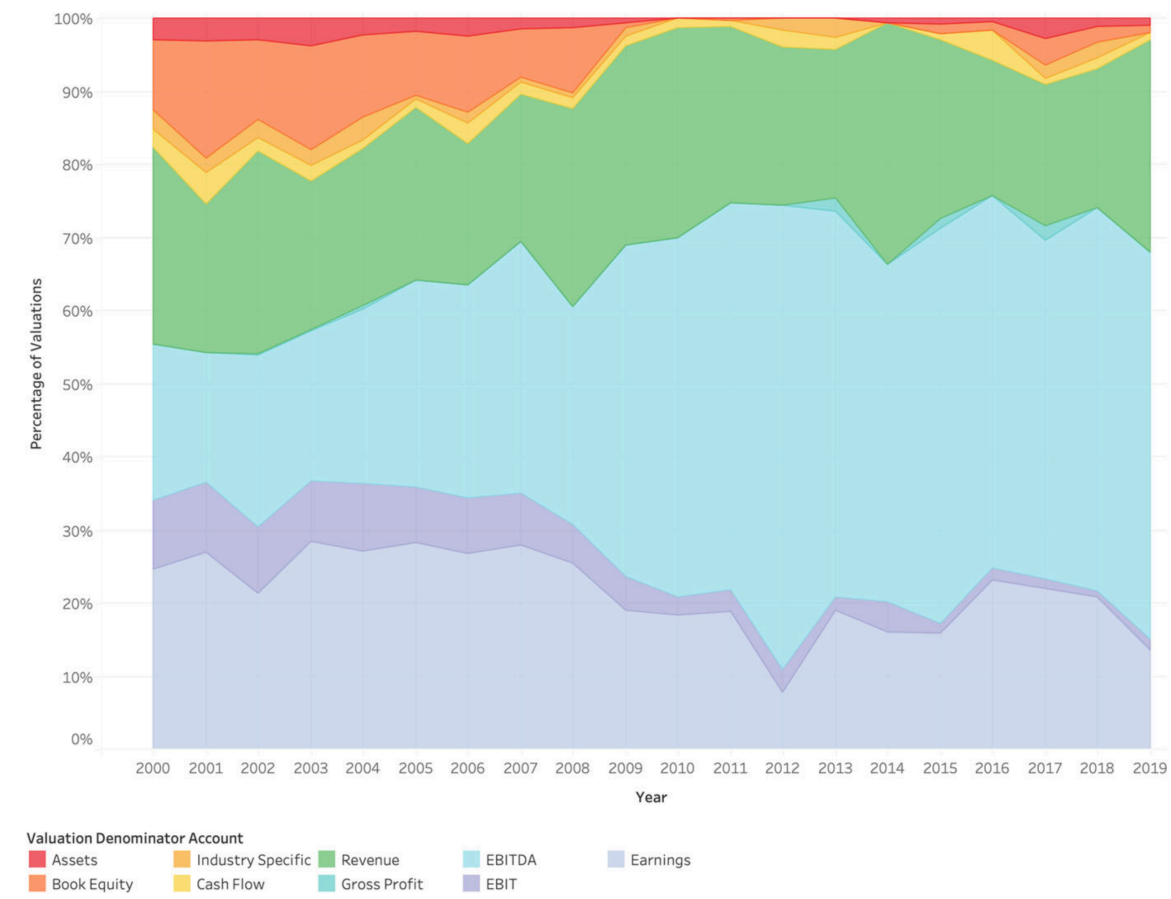


Fig. 3 Value-driver denominators over time. Figure 3 plots the relative frequency of value-driver denominators used in the valuation multiples, from 2000 through 2019

Value-driver **denominators** over time (Shaffer 2024)

Principles of **Market-Based Methods**

Segway through the Valuation Jungle

Assumption

Value can be derived from the market value of a **comparable company**

Relative Valuation Intuition

$$\text{Multiple} = \frac{\text{What you pay (price)}}{\text{What you get (book)}}$$

⇒ **Willingness to pay each EUR of earnings, cash flow, or sales, ...**

Principles of Market-Based Methods

Segway through the Valuation Jungle

Concept

$$MV_T = (MV_C / VI_C) \times VI_T$$

Where:

- MV_T = Market value of target company
- MV_C = Market value of comparable company C
- VI_C = Value indicator for comparable company C
- VI_T = Value indicator for target company T
- (MV_C / VI_C) = **Market value multiple**

⚠ P/E ratio equivalence from earlier: $P = \text{PEs of Peers} \times E$

Key Points

- Use multiples from comparable companies (peer firms)
- Multiples must be defined identically for all peers
- Select peers: similar profitability, risk, and growth characteristics
- Correlations help identify truly comparable peers (**Fairness Opinions!**)

Principles of Market-Based Methods **Example**

Segway through the Valuation Jungle

Valuing *Repsol YPF* Using Comparable Integrated Oil Companies

Comparable Company	Trailing P/E	Forward P/E	Price/Sales	Price/Book	Average
Exxon Mobil Corp. (XOM)	11.25	8.73	1.17	3.71	
British Petroleum (BP)	9.18	7.68	0.69	2.17	
Chevron Corp. (CVX)	10.79	8.05	0.91	2.54	
Royal Dutch Shell (RDS-B)	7.36	8.35	0.61	1.86	
ConocoPhillips (COP)	11.92	6.89	0.77	1.59	
Total SA (TOT)	8.75	8.73	0.80	2.53	
Eni SpA (E)	3.17	7.91	0.36	0.81	
PetroChina Co. (PTR)	11.96	10.75	1.75	2.10	
Average Multiple	9.30	8.39	0.88	2.16	
×					
Repsol YPF (Projections)	\$4.38	\$3.27	\$92.66	\$26.49	
= Estimated Market Value	\$40.72	\$27.42	\$81.77	\$57.32	\$51.81

Can you calculate a (Trailing) P/E ratio for a company? Try [eBay](#)!

Principles of Market-Based Methods **Example**

Segway through the Valuation Jungle

Valuing Repsol YPF Using Comparable Integrated Oil Companies

- Repsol has economic and political risks and growth characteristics similar to other globally diversified integrated oil and gas companies.
- The estimated value of Repsol based on the comparable companies method is \$51.81 billion versus its actual June 25, 2008, market capitalization of \$49.83 billion.
- The analyst needs to be mindful of changes in fundamentals that can affect multiples.
 - These fundamentals include a firm's ability to generate and grow earnings and cash flow through reinvestment in its operations, as well as the risk associated with the firm's earnings and cash flows.
 - Since multiples are affected by each of these variables, changes in the variables affect them. Firms with lower earnings and cash-flow generation potential, lower growth prospects, and higher risk should trade at multiples less than firms with higher earnings and cash-flow generation capability, higher growth prospects, and less risk.
 - Consequently, the analyst needs to understand why one firm's multiple is less than a comparable firm's before concluding that it is under- or overvalued.
 - **For example, a firm with a P/E of 10 may not be more expensive than a comparable firm with a P/E of 8 if the former's growth prospects, profitability, and rate at which profits are reinvested in the firm are higher than the latter firm's.**

Variations of Market-Based Methods

Segway through the Valuation Jungle

Recent Comparable Transactions

- Value of target based on purchase price of recently acquired comparable firms
- Calculate multiples for comparable transactions and apply to target
- Most accurate when transaction is truly comparable and very recent
- Includes a purchase price premium
- Truly comparable recent transactions are rare!

Same or Comparable Industry

$$MV_T = (MV_{IND} / VI_{IND}) \times VI_T$$

- Simplifies multiple selection by using industry average multiples
- Assumes industry multiples are comparable
- Requires addition of purchase price premium

Market-Based Methods

Segway through the Valuation Jungle

Pros of Relative Valuation

- Straight forward to apply
- Transparent assumptions
- Easy to explain
- Conceptual link with discount valuation models

Cons of Relative Valuation

- Are comparable firms correctly valued?
- Industry averages may not be good benchmarks (Apple's peer group includes Nokia and Motorola)
- How to deal with loss making/unlisted firms?
- Adjustment for a purchase price premium (except for recent transactions)
- Accounting distortions (accounting practices such as depreciation/amortization vs. expensing of R&D)

Segway through the Valuation Jungle

Valuation and Pricing

Selected Valuation Approaches

- Comparable firms, transactions ✓
- Enterprise value (EV) to EBITDA
- Price / Earnings to Growth ratio (PEG)
- Value-Driver-Based Valuation
- [...]

Multiples Scheme in Shaffer (2024)

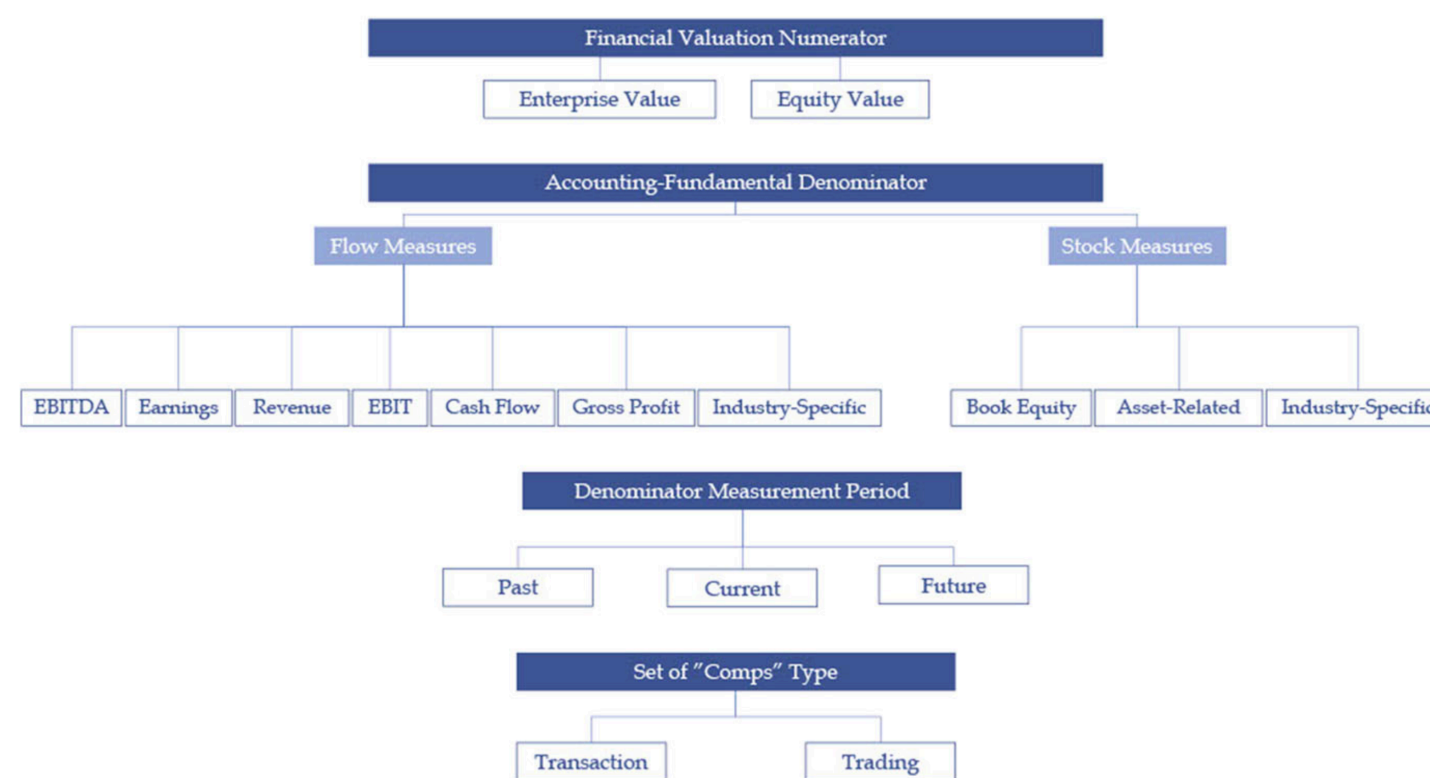


Fig. 1 Standardization scheme. Figure 1 illustrates the dimensions used to categorize the valuation ratios

Enterprise Value to EBITDA

Segway through the Valuation Jungle

$$EV/EBITDA = (MV_{FCFE} + MV_{PF} + (MV_D - \text{Cash}))/EBITDA$$

Where:

- MV_{FCFE} = Market value of equity
- MV_{PF} = Market value of preferred stock
- MV_D = Market value of debt
- $Cash$ = Cash and cash equivalents
- $EBITDA$ = Earnings before interest, taxes, depreciation, and amortization
- **EV = Enterprise Value**

→ *construct a multiple for the whole enterprise (equity and debt) rather than just equity*

Enterprise Value to EBITDA

Segway through the Valuation Jungle

Why useful?

- Capital structure → numerator is market value of **equity and debt** and denominator is operating income
 - Reflects obligation of acquirer to **pay off debt** (what it “costs” to buy the whole business)
 - Allows to compare firms with different levels of debt
 - Net of non-operating asset **cash** whose value is implicitly in equity value
- **EBITDA** less likely to be negative than earnings (no P/E ratio for loss-making firms)
- Limited sensitivity to accounting distortions (“before DA”)

Enterprise Value to EBITDA

Segway through the Valuation Jungle

Limitations

- Ignores impact of new investments on future cash flows (but firms can grow!)
- Often used for mature firms with stable cash flows and low growth rates

Enterprise Value to EBITDA **Example**

Segway through the Valuation Jungle

Repsol and Eni are geographically diversified integrated oil and gas companies. The market value of Repsol's common equity is \$40.36 billion, and Eni's is \$54.30 billion. Neither firm has preferred stock outstanding. Repsol's (\$14.6 billion) and Eni's (\$8.8 billion) outstanding debt consists primarily of interest-only notes (\$0.7 billion and \$0.3 billion, respectively) with a balloon payment at maturity. The average maturity date for Repsol's debt is 12 years; for Eni's debt it is 10 years. Market rates of interest for firms like Eni and Repsol at that time for debt maturing within 10 to 12 years were 7% and 7.5%, respectively.

Calculate EV/EBITDA ratios for Repsol and Eni.

Enterprise Value to EBITDA **Example**

Segway through the Valuation Jungle

$$EV/EBITDA = (MV_{FCFE} + MV_{PF} + (MV_D - \text{Cash}))/EBITDA$$

Ingredients

- MV_{FCFE} → “The market value of Repsol’s common equity is \$40.36 billion, and Eni’s is \$54.30 billion”
- MV_{PF} → “Neither firm has preferred stock outstanding.”
- MV_D → ?
- $Cash$ → ? [additional info needed]
- $EBITDA$ → ? [additional info needed]

Enterprise Value to EBITDA **Example**

Segway through the Valuation Jungle

MV (= Present Value) of Repsol's Long-Term Debt

Use the following formula to calculate the market/present value of the annuity payment

$$\text{MV Annuity} = \text{Payment} \times \frac{1 - (1 + i)^{-n}}{i}$$

then **add** the present value of the balloon payment at maturity

$$\text{MV Balloon} = \frac{\text{Balloon Payment}}{(1 + i)^n}$$

to derive at the market value of debt:

$$\text{MV}_D = \$0.7 \times \left[\frac{1 - 1/(1.075)^{12}}{0.075} \right] + \frac{\$14.6}{1.075^{12}} = \$0.7 \times 7.74 + \$6.13 = \$11.55\text{bn}$$

Enterprise Value to EBITDA **Example**

Segway through the Valuation Jungle

MV of Eni's Long-Term Debt

$$MV_D = \$0.3 \times \left[\frac{1 - 1/(1.070)^{10}}{0.07} \right] + \frac{\$8.8}{1.07^{10}} = \$0.3 \times 7.02 + \$4.47 = \$6.58\text{bn}$$

i Book value of debt is often used as a proxy for market value, but this can be misleading for long-term or off-market debt.

Enterprise Value to EBITDA **Example**

Segway through the Valuation Jungle

$$EV/EBITDA = (MV_{FCFE} + 0 + (MV_D - \text{Cash}))/EBITDA$$

Balance Sheet	Repsol	ENI
Cash	3.8	6.2
...		
Income Statement	Repsol	ENI
Depreciation	4.10	8.10
EBIT	8.0	27.6
Interest expense	0.7	0.3
...		

Repsol

$$\frac{(\$40.36 + 0 + \$11.55 - \$3.8)}{(\$8.0 + \$4.1)} = 3.98$$

Eni

$$\frac{(\$54.30 + 0 + \$6.58 - \$6.2)}{(\$27.60 + \$8.1)} = 1.53$$

Segway through the Valuation Jungle

Valuation and Pricing

Selected Valuation Approaches

- Comparable firms, transactions ✓
- Enterprise value (EV) to EBITDA ✓
- Price / Earnings to Growth ratio (PEG)
- Value-Driver-Based Valuation
- [...]

Multiples Scheme in Shaffer (2024)

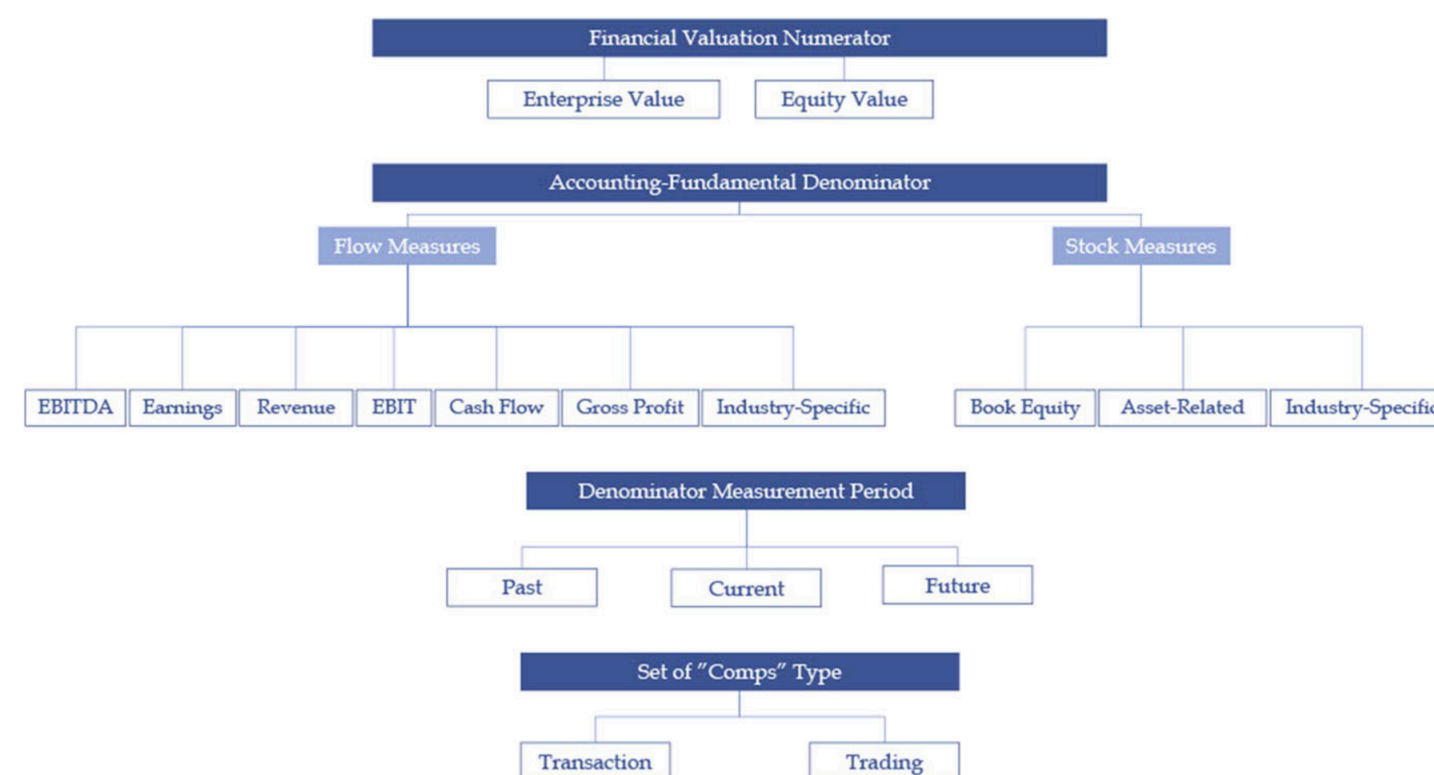


Fig. 1 Standardization scheme. Figure 1 illustrates the dimensions used to categorize the valuation ratios

Adjusting Relative Valuation: **PEG Ratio**

Segway through the Valuation Jungle

Assume that 2 firms are direct competitors and have P/E ratios of 10 and 20, respectively.

Which is the cheaper firm?

→ Not possible to answer without knowing growth rates of earnings (and what return can be earned from reinvested earnings)!

→ The **PEG ratio adjusts for differences in growth rates** among comparable firms

Adjusting Relative Valuation: **PEG Ratio**

Segway through the Valuation Jungle

→ The **PEG ratio (A)** adjusts for differences in growth rates among comparable firms:

$$A = \frac{MV_C / VI_C}{VI_{CGR}}$$

Where:

- MV_C / VI_C = Market price to value indicator multiple (P/E) for comparable firms
- VI_{CGR} = Projected growth rate in value indicator for comparable firms

⇒ **A is the PEG ratio for comparable firms**

Adjusting Relative Valuation: **PEG Ratio**

Segway through the Valuation Jungle

When applying this adjustment, the Market Value Multiple Valuation

$$MV_T = (MV_C / VI_C) \times VI_T$$

becomes

$$MV_T = A \times VI_{TGR} \times VI_T$$

where

- A = **PEG ratio for comparable firms**, which is the market price to value indicator relative to growth rate for comparable firms
- VI_T = Value indicator for target
- VI_{TGR} = Projected growth rate in value indicator of target firm

Adjusting Relative Valuation: PEG Ratio **Example**

Segway through the Valuation Jungle

An analyst is asked to determine whether Basic Energy Service (BES) or Composite Production Services (CPS) is more attractive as an acquisition target. Both firms provide engineering, construction, and specialty services to the oil, gas, refinery, and petrochemical industries. BES and CPS have *projected annual earnings per share growth rates of 15% and 9%, respectively*. BES's and CPS's *current earnings per share are \$2.05 and \$3.15, respectively*. The current *share prices for BES are \$31.48 and for CPX are \$26.00*. The industry average [assume that industry peers are comparable firms here] *price-to-earnings ratio and growth rate are 12.4% and 11%, respectively*.

Calculate valuations using the PEG ratio (for comparable firms) and discuss which firm is a more attractive takeover target as of now.

Adjusting Relative Valuation: PEG Ratio **Example**

Segway through the Valuation Jungle

Calculate Valuations Using PEG Ratio

- Industry average PEG ratio: $A = \frac{MV_C / VI_C}{VI_{CGR}} = \frac{12.4}{11} = 1.13$
- BES implied share price: $MV_T = A \times VI_{TGR} \times VI_T = 1.13 \times 15 \times 2.05 = 34.66$
- CPS implied share price: $MV_T = A \times VI_{TGR} \times VI_T = 1.13 \times 9 \times 3.15 = 31.96$
- Important: Consistent percentage point vs. decimals in growth rates!

Adjusting Relative Valuation: PEG Ratio **Example**

Segway through the Valuation Jungle

Takeover Target Attractiveness

Firm	Implied Price	Current Price	Upside (%)
BES	\$34.66	\$31.48	+10.1%
CPS	\$31.96	\$26.00	+23.0%

- CPS appears to be a more attractive acquisition target than BES based on the PEG ratio, as it has a higher upside potential of 23% compared to BES's 10.1% (although BES has both a higher growth rate and a higher implied share price).
- BES might be a more attractive acquisition target if it generates increasing future financial returns (and if its projected earnings stream is viewed as less risky).

Adjusting Relative Valuation: PEG Ratio

Segway through the Valuation Jungle

Firm PEG Ratio

One can move from a PEG ratio for comparable firms to a PEG ratio for the target firm by using (i) the target firm's market price to value indicator in the numerator (MV_T/VI_T , e.g., P/E) and (ii) the target firm's projected growth rate (VI_{TGR}) in the denominator:

$$A_T = \frac{MV_T/VI_T}{VI_{TGR}}$$

Interpretation

- $A_T > A \rightarrow$ target priced more expensively than peers (conditional on growth)
- $A_T < A \rightarrow$ target priced more cheaply than peers (conditional on growth)

Segway through the Valuation Jungle

Valuation and Pricing

Selected Valuation Approaches

- Comparable firms, transactions ✓
- Enterprise value (EV) to EBITDA ✓
- Price / Earnings to Growth ratio (PEG) ✓
- Value-Driver-Based Valuation
- [...]

Multiples Scheme in Shaffer (2024)

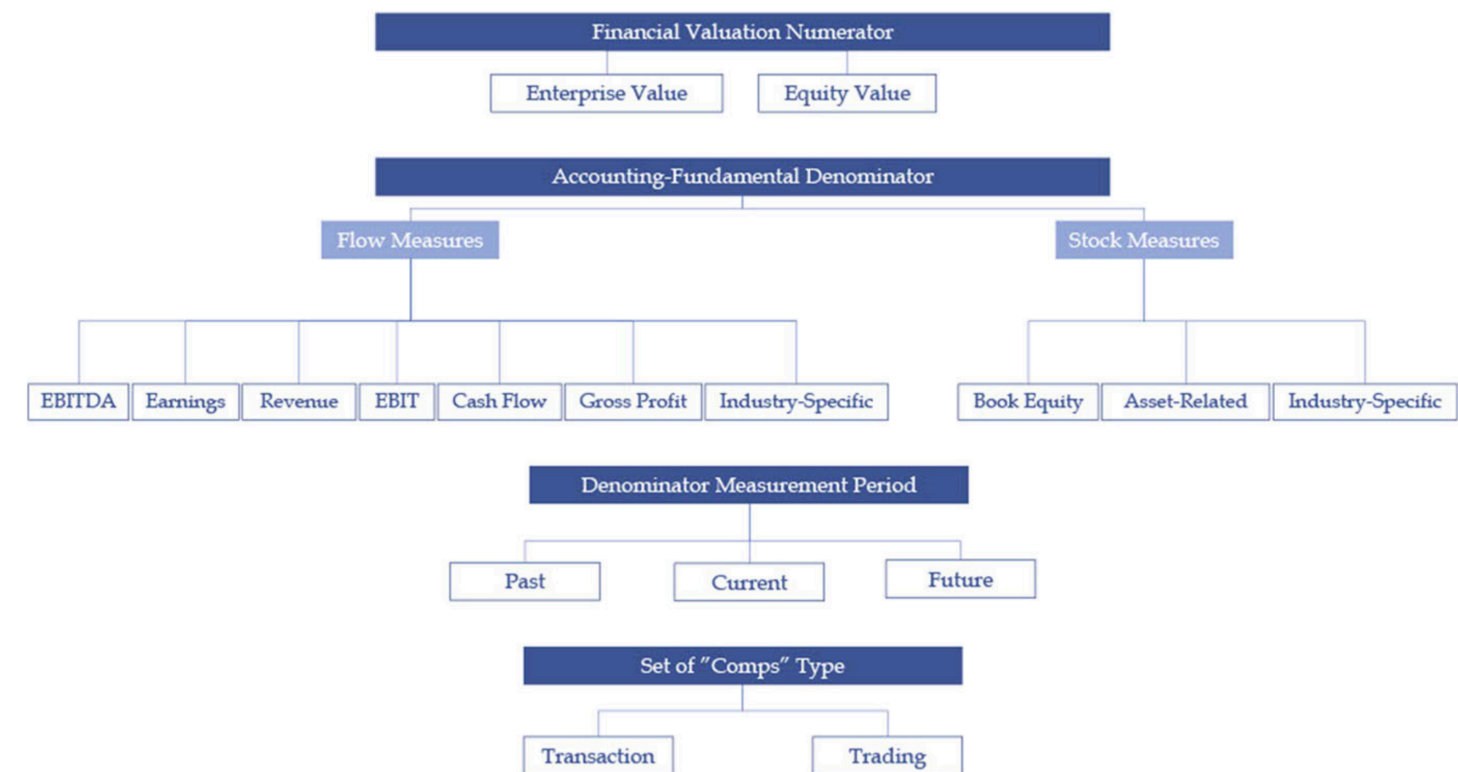


Fig. 1 Standardization scheme. Figure 1 illustrates the dimensions used to categorize the valuation ratios

Value-Driver-Based Valuation

Segway through the Valuation Jungle

What to do when earnings are missing?

Micro Value Drivers

- Influence **specific functions** within the firm, e.g., sales, marketing, production, and distribution (e.g., Part Defects per X Units Sold)

Macro value drivers

- Influence all aspects of the firm (e.g., *market share* or overall customer satisfaction)

Value-Driver-Based Valuation

Segway through the Valuation Jungle

Applying Value-Driver-Based Valuation

1. Identify value drivers of the firm
 2. Market value for comparable companies divided by value driver = \$ of market value per unit of value driver
 3. Multiply with same value driver for target firm = MV_T
- ⇒ implied assumption that a **single value driver is representative** of the total value of the business (Dot-com bubble)

Segway through the Valuation Jungle

Valuation and Pricing

Selected Valuation Approaches

- Comparable firms, transactions ✓
- Enterprise value (EV) to EBITDA ✓
- Price / Earnings to Growth ratio (PEG) ✓
- Value-Driver-Based Valuation ✓
- [...] → DePamphilis (2025)

Multiples Scheme in Shaffer (2024)

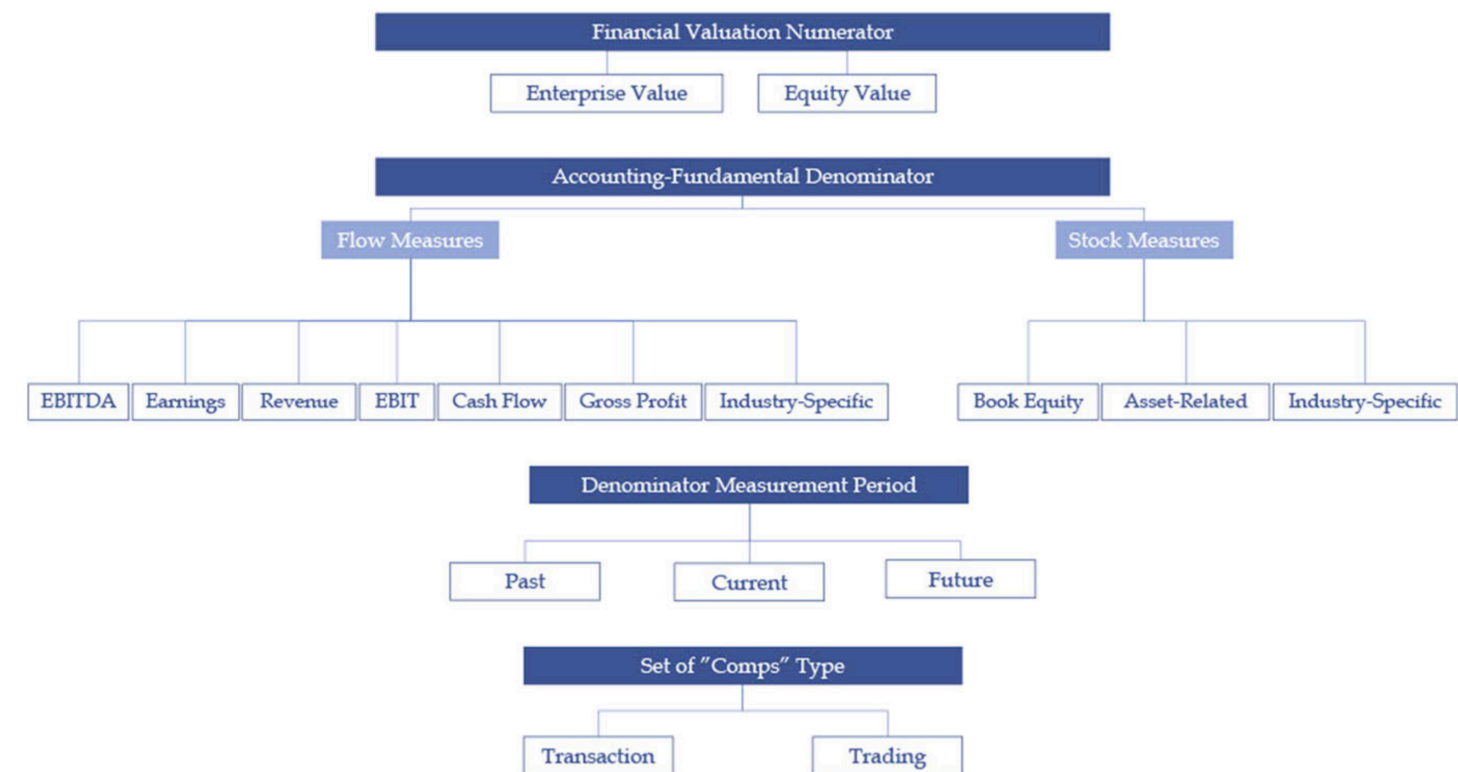


Fig. 1 Standardization scheme. Figure 1 illustrates the dimensions used to categorize the valuation ratios

When to use which method?

Methodology	Use this methodology when:
Discounted cash flow	• The firm is publicly traded or private with identifiable cash flows • A start-up has some history of facilitating cash flow forecasts • An analyst has a long-time horizon • An analyst has confidence in forecasting the firm's cash flows • Current or near-term earnings or cash flows are negative but are expected to turn positive in the future • A firm's competitive advantage is expected to be sustainable • The magnitude and timing of cash flows vary significantly
Comparable companies	• There are many firms exhibiting similar growth, return, and risk characteristics • An analyst has a short-term time horizon • Prior, current, or near-term earnings or cash flows are positive • An analyst has confidence the markets are, on average, right • Sufficient information to predict cash flows is lacking • Firms are cyclical; for P/E ratios, use normalized earnings (i.e., earnings averaged throughout the business cycle) • Growth rate differences among firms are large; use the PEG ratio
Comparable transactions	• Recent transactions of similar firms exist • An analyst has a short-term time horizon • An analyst has confidence the markets are, on average, right • Sufficient information to predict cash flows is lacking
Same or comparable industry	• Firms within an industry or a comparable industry are substantially similar in terms of profitability, growth, and risk • An analyst has confidence the markets are, on average, right • Sufficient information to predict cash flows is lacking
Replacement-cost approach	• An analyst wants to know the current cost of replicating a firm's assets • The firm's assets are easily identifiable, tangible, and separable • The firm's earnings or cash flows are negative
Tangible book value	• The firm's assets are highly liquid • The firm is a financial services or product distribution business • The firm's earnings and cash flows are negative
Breakup value	• The sum of the value of the businesses/product lines comprising a firm is believed to exceed its value as a going concern
Liquidation value	• An analyst wants to know asset values if they were liquidated today • Assets are separable, tangible, and marketable • Firms are bankrupt or subject to substantial financial distress • An orderly liquidation is possible
Real options (contingent claims)	• Additional value can be created if management has a viable option to expand, delay, or abandon an investment • Assets not currently generating cash flows have the potential to do so • Assets have characteristics most resembling financial options • The asset owner has some degree of exclusivity (e.g., a patent)

Apples and Peers: Empirical Evidence

Comparable Companies

Peer Selection and Valuation in M&A (Eaton et al. 2022)

Research Question

What drives investment IBs' choice of peers in comparable companies analysis?

Results

- **Product market space** is among the most important factors in peer selection
 - but: **SIC codes do a poor job** of categorizing related firms
- Banks strategically select large, high growth peers with high valuation multiples:
 - helps target to negotiate higher takeover price
 - against banks' incentives to collect success fees (lower valuations could increase deal closing probability)

UBS's opinion: Schneider Electric SA buys BEI Technologies

Peer Selection and Valuation in M&A ([Eaton et al. 2022](#))

Selected Companies Analysis

UBS reviewed and compared certain financial information for the Company to corresponding financial information, ratios and public market multiples for the following selected publicly traded industrial technology companies:

Automotive Concentration

- Gentex Corporation
- Melexis Microelectronic Systems
- Stoneridge, Inc.
- TT Electronics plc

Industrial Technology Concentration

- Amphenol Corporation
- CTS Corporation
- Molex, Inc.
- Rofin-Sinar Technologies, Inc.

Aerospace and Defense Concentration

- EDO Corporation
- Esterline Technologies

Although the selected companies were used for comparison purposes, the businesses of no **selected company** were either **identical or directly comparable** to the businesses of the Company. Accordingly, UBS' comparison of the selected companies to the Company and analysis of the results of such comparisons was **not purely mathematical**, but instead necessarily involved **complex considerations** and judgments concerning differences in **financial and operating characteristics** and other factors that could affect the relative values of the selected companies and of the Company.

Takeaways

Fairness Opinions

- Board's Fiduciary Responsibilities (duties of care and loyalty) in transactions require professional/external evaluation of transaction fairness
- FOs historically evolved following the Trans Union case
 - Can be strategic tools influenced by litigation risk and agency conflicts
 - Fee structures create potential conflicts of interest

Measuring Value

- Market-based methods are derived from fundamental discount principles
- Multiples (comparables, transactions, asset-based) serve different contexts
- Method selection depends on company characteristics and available information
- Each method has strengths and limitations

Thank You!

RSM

Ezra

References

- DePamphilis, Donald M. 2025. *Mergers, Acquisitions, and Other Restructuring Activities*. Academic Press.
- Eaton, Gregory W, Feng Guo, Tingting Liu, and Micah S Officer. 2022. “Peer Selection and Valuation in Mergers and Acquisitions.” *Journal of Financial Economics* 146 (1): 230–55.
- Shaffer, Matthew. 2024. “Which multiples matter in M&A.” *Review of Accounting Studies* 29 (3): 2724–52.

Copyright

© Erasmus University Rotterdam, all rights reserved. No text and data mining.

English

This study material is protected by copyright under Dutch copyright laws. It is provided solely for your personal use and study. You are allowed to create a copy for your personal use. You are not allowed to make this study material or parts thereof available to others or reproduce it without prior written permission of the Erasmus University Rotterdam.

Nederlands

Dit onderwijsmateriaal is een auteursrechtelijk beschermd werk in de zin van de Auteurswet. Het wordt in het kader van uw studie aan u ter beschikking gesteld en is uitsluitend bedoeld voor uw eigen persoonlijke gebruik. U mag daarvoor een persoonlijke kopie maken. Het verveelvoudigen en openbaar maken (waaronder begrepen het aan derden ter beschikking stellen en/of verspreiden) van dit onderwijsmateriaal of een gedeelte daarvan is niet toegestaan zonder voorafgaande schriftelijke toestemming van de Erasmus Universiteit Rotterdam.

Titlepage photo by [Dennis Möller](#) on [Unsplash](#)